Muhammad Haseeb

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RESEARCH INTERESTS

Mechanistic Interpretability: Analyzing internal circuit evolution for domain generalization and to understand representational collapse in Federated Learning.

Domain Generalization: Adapting Models to distribution shifts through pruning to identify sparse subnetworks that preserve domain-invariant representations.

Efficient ML: Compression methods (pruning, quantization) to save compute and memory for LLMs without loss in performance.

EDUCATION

Lahore University of Management Sciences (LUMS) Sep 2022 – May 2026
B.S. Computer Science
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, AI on Edge Devices, Advanced Topics in ML, LLM Systems, Linear Algebra, Probability.

PUBLICATIONS

- **Mechanistic Evidence for Preserved-but-Misaligned Representations in Non-IID FedAvg (*ICML 2026 Mechanistic Interpretability Workshop*).**
  Muhammad Haseeb, Salaar Masood, Muhammad Abdullah Sohail, Mohammad Fatim Shoaib, Muhammad Tahir.
- **BaCP: Backbone Contrastive Pruning for Preserving Representations in Extremely Sparse Neural Networks (*Preprint*).**
  Mohammad Haroon Khawaja, **Muhammad Haseeb**, Mohammad Fatim Shoaib, Muhammad Tahir.

RESEARCH EXPERIENCE

ML Research Assistant Jan 2025 – Present
Centre for Urban Informatics, Technology and Policy (CITY), LUMS
Advisors: Dr. Muhammad Tahir, Dr. Zubair Khalid

- **Mechanistic Analysis of Federated Learning:** (*Accepted at ICML 2026 Mechanistic Interpretability Workshop*)
  - Studied FedAvg through the lens of Mechanistic Interpretability to analyze performance loss under Non-IID conditions using class-specific circuit discovery, Universal Sparse Autoencoders (USAEs), and linear probes.
  - Compared class-specific circuits and observed drift and interference patterns during non-IID aggregation, while IID circuits behaved in a perfect constructive manner.
  - Demonstrated via USAE concept analysis that useful latent features, similar to the IID feature basis, remain present and recoverable even when global accuracy degrades under label skew.
- **Backbone Contrastive Pruning (BaCP):**
  - Contributed to a generalized pruning framework to merge standard pruning criteria with contrastive learning to prevent representational collapse.
  - Designed a multi-objective loss function that aligns sparse embeddings with pretrained, fine-tuned, and historical snapshot references to maintain feature consistency with pretrained references to prevent representational collapse.
  - Maintained accuracy comparable to dense baselines at up to **99% sparsity**, outperforming standard unstructured pruning approaches across ViTs/CNNs/Language Models.
- **Domain Generalization & Interpretability:**
  - Applied circuit extraction techniques to isolate task-specific subnetworks, analyzing whether out-of-distribution failures are caused by shortcut learning or weight interference.
  - Optimized visual queries in Vision Transformers using reinforcement learning (Group Relative Query Opti-

mization), achieving a **3% accuracy gain** on the PACS dataset compared to empirical risk minimization.

- **Quantization of Diffusion Models:**
  - Proposed **Log-SNR Conditioned Temporal Dynamic Quantization** to handle non-stationary activations, achieving a **55% reduction in LPIPS** on PixArt-α.
  - Implemented Hessian-based optimization (GPTQ/QroNos) for **4-bit weight compression**, utilizing second-order curvature information to mitigate quantization errors.

AI/SWE Research Intern May 2025 – Aug 2025
University of Illinois Urbana-Champaign (UIUC)

- Investigated LLM-based heuristics for automated `#ifdef` guard insertion in C codebases and explored large language models for software debloating and code dependency analysis.
- Contributed to the development of a VS Code extension supporting a C-to-Rust translation pipeline.

ML Research Assistant Jan 2025 – Aug 2025
Computer Vision & Graphics Lab, LUMS

- **KL Aware Quantization (KLAWQ):** Proposed an augmented GPTQ framework integrating KL divergence and second-order curvature to optimize a combined MSE+KL objective, reducing LLM perplexity by **30%** compared to GPTQ baselines.
- **Single-Image Camera Calibration:** Developed a pipeline to train a Multi-Scale Deformable Transformer by combining loss with geometric constraints to recover full camera projection matrices from a single image.

INDUSTRY EXPERIENCE

Founding Product Engineer June 2026 – Present
NEXA (Remote)

- Architecting and building highly scalable, distributed backend systems from the ground up.
- Integrating machine learning models and optimizing distributed serving pipelines for fast, efficient inference.

OSS Engineer Dec 2025 – Feb 2026
DatacurveAI (YC S24)

- Solved bugs and added new features across various open-source machine learning repositories to test and train LLMs on working with large codebases.

Machine Learning Engineer Jul 2025 – Oct 2025
Innova Tech

- Automated data annotation pipeline, refined training configurations, and fixed model architectures getting a **4% accuracy gain**. Also, implemented inference optimization with TensorRT for speedup and various quantization techniques to reduce memory footprint by **40%** on edge devices.

TEACHING EXPERIENCE

Teaching Assistant Sep 2025 – Dec 2025
CS436: Computer Vision, LUMS

- Designed and supervised an assignment on transfer learning and multithreaded C++ object detection. Mentored 40 student groups in building a virtual tour application using Structure from Motion.

SKILLS & OPEN-SOURCE

Technical Skills: Python, C, C++, Rust, SQL; PyTorch, Transformers, ONNX, TensorRT, diffusers, adapters.

Open-Source Contributions:

- **pytorch-image-models:** Added F1, precision, and recall metrics to evaluation and training pipelines.
- **adapters:** Implemented PEFT support for Group Query Attention models, fixing tensor mismatch issues.